Listening with a Mealworm Beetle: The Contact Membrane and the Politics of Umwelt-centric Design

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Figure 1: Mealworm beetle and human performing through the Contact Membrane

Abstract

Interfaces staging non-human life often prioritize access, rendering it legible or simulating its viewpoint. This paper introduces the Contact Membrane, an interface for real-time listening with a mealworm beetle, and its method, Umwelt-centric design. A tensioned paper surface, sensed by contact microphones and an IMU, turns the beetle's footsteps into a spatialized soundscape as a performer tilts it to keep the insect aloft. Across three performances, its beauty drew audiences in, yet made some doubt whether it masked the beetle's experience. That doubt is the contribution: Umwelt-centric design surfaces the politics of interspecies listening—who is heard, on whose terms.

Keywords

Bio-hybrid Interfaces, Animal-Computer Interaction, Sonification, Tangible Interfaces, Research through design

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1 Introduction

Relational aesthetics reframed the artwork as the production of relations rather than objects[Bourriaud 1998], yet those relations have largely remained bound to a limited inter-human social sphere[Bishop 2004], excluding many humans and non-humans. While the more-than-human turn extends this relationality beyond the human [Haraway 2013], technological engagements with non-human life have largely kept the organism at a distance—observed, displayed, or sensed at the scale of environments. What remains less explored is intimate, real-time, reciprocal listening on a non-human subject's own perceptual terms, even as animal-centered design has been called for [Mancini 2011].

Contact Membrane is an interface and an artwork that stages a real-time relationship with a single mealworm beetle, developed by a research through design process. Through the process we asked how technology can mediate interspecies communication with an emphasis on listening. In this paper, we propose a design method we call Umwelt-centric design, and report findings suggesting that interfaces built through it can serve as a speculative means of surfacing the politics of interspecies listening.

2 Related works: More-than-human interfaces

Recent surveys on bio-hybrid interfaces reveal that despite the theoretical shift toward non-human actors, design practices remain largely anchored to anthropocentric paradigms [Kneile et al. 2025].

Organisms are frequently framed as subjects to be revealed, explored, or made completely legible. This drive for legibility aligns with what Haraway terms the "optical regime"—a visual dominance that fixes the other as an image to be known, categorized, and mastered [Haraway 1988]. To engage with non-human agency without reducing it to a visible object, we propose a methodological shift toward listening. Listening is a mode that allows sound to arrive rather than being actively grasped or possessed [Nancy and Mandell 2009], and it can be an effective way of framing an interaction. However, it is worth noting that listening: like vision, is mediated, and that mediation carries its own politics.

In the context of interactive art, sound is a discipline that focuses on listening. However, integration of biological systems into sonic media has historically prioritized human control and instrumentalization over listening. This tendency spans various approaches: pioneering works like Alvin Lucier's Music for Solo Performer relied on a concealed chain of technical curation to fit a specific musical aesthetic [Straebel and Thoben 2014]; interfaces such as Botanicus Interacticus reduced biological organisms to capacitive controllers for human gestures [Poupyrev et al. 2012]. A recent line of work in this venue simulates the non-human point of view through sensory substitution, letting a human momentarily become another organism—echolocating like a bat or sensing proximity like a mole (e.g., [Hu et al. 2024; Huang et al. 2025]). While these works productively unsettle human sensory defaults, they operate by acquiring the other's perspective: the organism itself is absent, modeled rather than met. Our approach instead stays with a living individual—it does not attempt to possess or inhabit the insect's subjectivity, but foregrounds the difference and the political friction of the encounter itself.

3 Concept and Theoretical Framework

3.1 Designing interspecies interfaces

Following the biologist Jakob von Uexküll, every organism inhabits its own Umwelt: a self-world carved out of the environment by the specific perceptual and effector capacities of its body [von Uexküll and O'Neil 2013]. If we are to listen to a more-than-human other intimately as a subject rather than an object, we must orient toward its perceptual world. Doing so demands technological mediation between two fundamentally different perceptual systems. Such mediation, however, is never neutral: as Ihde's postphenomenology argues, technology actively constructs human perception [Ihde 1990], and media studies likewise hold that what we perceive, and what we fail to notice is shaped by culture[Sterne 2003]. The media through which we engage insect-scale organisms must therefore be designed critically.

Because mediation is never neutral, translating between Umwelten is a political action: every design decision curates whose perception is prioritized and determines exactly how a non-human body is rendered sensible to human ears. In this sense the interface enacts what Rancière calls a distribution of the sensible: a partition of what can be heard and what remains inaudible [Rancière 2004]. By "political" we mean precisely this power-laden curation. In this paper, we propose Umwelt-centric design as a response: a methodology that structures the interface around the sensory modalities of the non-human subject while explicitly owning the imperfection of any such translation.

Table 1: Instrumental mapping versus Umwelt-centric mapping.

	Instrumental	Umwelt-centric
Orientation	Control and legibility	Listening on the subject's terms
Signal	Movement / control data	Vibration (the subject's channel)
Noise	Filtered for clarity	Preserved as meaningful
Time	Human / musical tempo	Species-specific (granular)
Role	Operator	Listener

3.2 Beyond instrumentation: The artwork as process

The development of the Contact Membrane and the concept of Umwelt-centric design operated as a Research through Design process [Zimmerman et al. 2007], mobilizing Haraway's notion of "becoming-with" as an active design methodology [Haraway 2013]. This approach is best illustrated by a critical pivot in the project's early stages. The interface was initially built as a small, insect-sized hammock that hosted the mealworm beetle, using accelerometers to translate the structure's macroscopic movements into sound. During a preliminary performance, however, observers raised pointed questions: Is the bug just a tool in this case? Is there a genuine connection, or is the interaction fundamentally one-sided? This critique exposed a contradiction: despite our intention of interspecies engagement, the interface still operated as an anthropocentric control mechanism, extracting data from the insect rather than relating to it.

Prompted by this finding, we made a material pivot to enact listening. To receive the mealworm beetle's presence as a subject, we replaced the hammock with a tensioned paper membrane designed to capture the microscopic acoustic friction of its footsteps. This move from movement-data to direct acoustic data transformed the relational dynamic. Hearing the mealworm beetle navigate the paper made its presence vibrant, and this drew the research closer to the mealworm beetle itself: toward rendering sound from their perspective as faithfully as possible, and sharing a soundscape with them. The iterative construction of the artwork was itself the practice of learning to listen, and of becoming-with the mealworm beetle.

Figure 2: The earlier iteration: an insect-sized hammock using accelerometers to translate movement into sound.

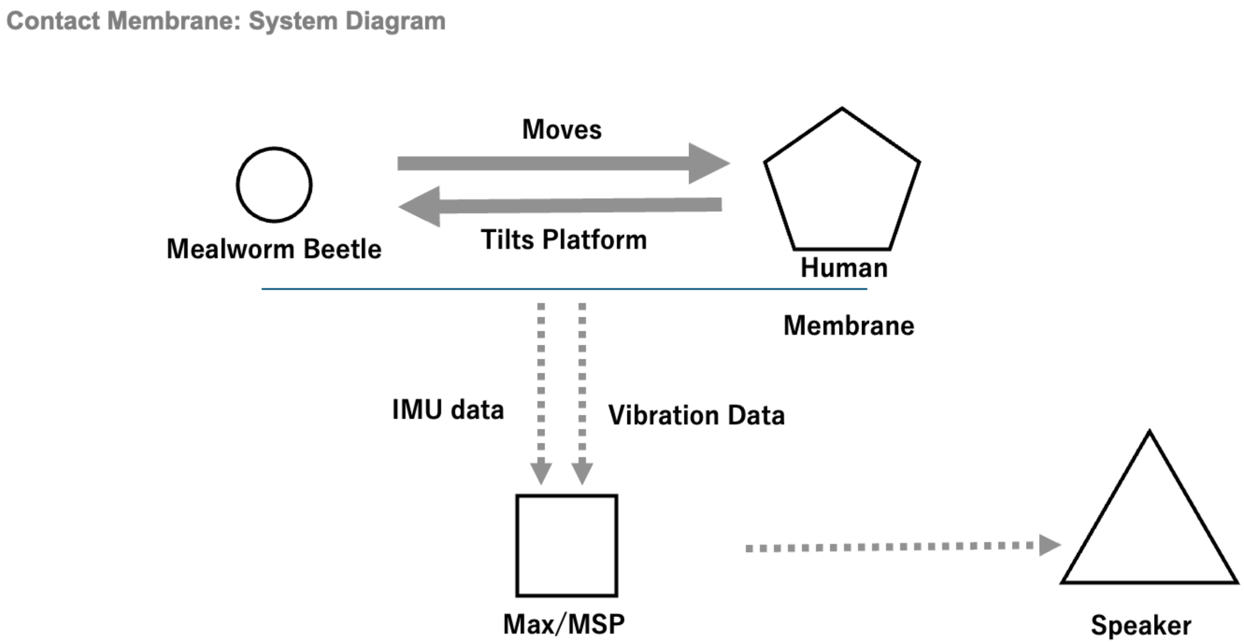

Figure 3: Detailed interaction loop between the mealworm beetle and the human performer.

4 The Contact Membrane: System Design

The Contact Membrane's system architecture comprises three integrated layers: a physical paper interface, a multi-modal sensing array (piezoelectric and IMU), and a real-time spatial audio processing engine built in Max/MSP.

4.1 Hardware Architecture

The core physical interface is a highly tensioned membrane derived from the surface layer of corrugated cardboard. This material was selected for its ability to translate vibration, which aligns with the mealworm beetle's Umwelt [Hill 2008]; and additionally, it allows the mealworm beetle to walk on it easily because it can grab it. Piezoelectric microphones capture these vibrational signals, while an Inertial Measurement Unit (IMU) monitors the device's spatial orientation. Within this hardware setup, the gravitational instability of the interface invites listening. Because the human must manually tilt the frame to prevent the insect from falling, this structural coupling compels the human to surrender absolute control.

4.2 Mapping data to sound

With the theoretical framework of Umwelt-centric design established in Section 3.1, the sensory data is mapped through three primary strategies: the preservation of noise, the translation of time scale, and spatialization.

First, sounds such as friction, scratching, and handling are preserved rather than masked. Because the beetle primarily experiences its environment through substrate vibration [Hill 2008], retaining these sonic cues is an implementation choice that reflects the experience of the mealworm beetle.

Second, the system operationalizes Uexküll's notion of the species-specific moment, in which time is more finely grained [von Uexküll and O'Neil 2013] for smaller beings, through a progression logic tied to structural stability. As the interaction progresses, the system introduces granular synthesis to stretch and magnify the microscopic transients of the beetle's footsteps. Stretching these transients makes them perceptible and playable for the human, while remaining faithful to the beetle's denser temporality. It must be noted that there is no one correct mapping under Umwelt-centric design, and here is where the political tension between aesthetics and faithfulness surfaces.

Finally, spatialization links the sound field to the physical act of balancing. The kinematic data from the IMU is mapped to sound-source positions within an Ambisonic environment, so that as the human tilts the frame to keep the beetle from falling, the amplified sounds pan across the audio field. This renders audible the gravity-bound situation the beetle inhabits that is otherwise imperceptible to the human, anchoring the listener's attention.

5 Performance and reflection

To evaluate the interface in a public context, we conducted three iterations of the performance within a university environment,

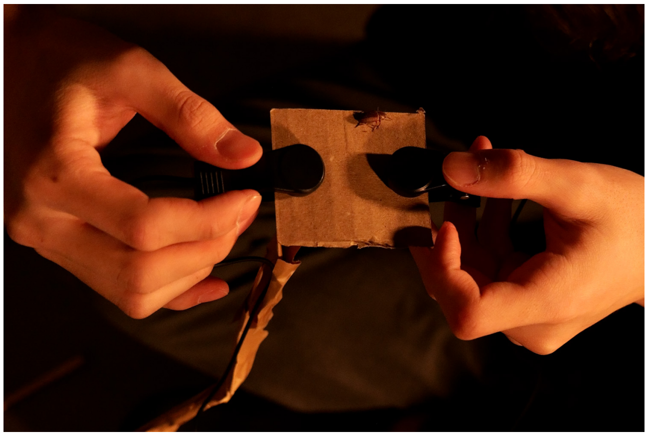

Figure 4: The performer tilting the membrane to keep the beetle aloft.

Figure 5: The performance setup: the tensioned paper membrane, contact microphones, IMU, and the Max/MSP spatial-audio chain that together compose the entanglement.

attended by audiences of approximately 15, 10, and 15 individuals, respectively. The interface was used to perform for around 10 minutes each, then some audiences experienced the interface themselves afterwards. Feedback was gathered through reflection from the performer, and unstructured oral interviews with a few audience members following each session.

5.1 Performer's perspective

The performer noted that the continuous necessity of attuning to the vibrations of the mealworm beetle during the performance resulted in a heightened sensitivity to micro-scale phenomena extending beyond the duration of the performance, into their daily lives.

5.2 Audience Feedback

Feedback from the audience centered on themes of intimacy, agency, and aesthetic tension. First, observing a live insect physically entangled with a human performer provoked direct ethical questions regarding the event itself. Audience members asked questions such as, "Is this a good experience for the insect?" "How are they feeling?". Second, while listeners often described the spatialized granular synthesis as "beautiful" or "immersive," some members questioned whether the beauty of the curated soundscape was masking the physical experience of the insect navigating the membrane.

6 Discussion: Aesthetics and politics in more-than-human art

Across the performances, the transformation of raw insect vibrations into a spatialized soundscape functioned as an aesthetic lure: its beauty drew listeners in and sustained their attention past an initial aversion to a live insect. Yet the same beauty provoked skepticism: several audience members asked whether the curated soundscape masked the beetle navigating the membrane. We read this doubt as a central achievement. The interface turns attention back onto the act of representation itself, in the manner of reflective design, whose aim is to provoke critical reflection on the assumptions a technology embeds rather than to disappear behind them [Sengers et al. 2005].

The soundscape is a curation rather than a recording. The design decisions required to render the micro-vibrations sensible, such as the choice of tensioned paper, microphone placement, and spatial processing, do not transmit a pre-existing "voice" of the mealworm beetle. The signal is an entanglement: it emerges through the intra-action of the insect's locomotion, the human's balancing, and the media, none of which holds the resulting agency in advance [BARAD 2007]. What the audience hears is a co-production, not an authentic interior.

Because this curation determines what of the beetle becomes audible, and on whose terms, it is political: a distribution of the sensible [Rancière 2004]. The contribution of Umwelt-centric design is that it does not hide this. By staging the translation as openly partial and letting its beauty become doubtable, the method works speculatively [Dunne and Raby 2013]: it surfaces the power asymmetries embedded in more-than-human interaction design rather than resolving or concealing them.

7 Limitations

We acknowledge three limitations. First, the work engages a single species, the mealworm beetle; whether Umwelt-centric design generalizes, particularly to organisms with very different Umwelten or bodily scales remains to be tested. Second, the evaluation is small and situated: to keep the handling of a live animal controlled, the author served as the main performer, and our reflections draw on informal feedback from small audiences across three sessions, so broader and more systematic evaluation with independent performers and participants is needed. Third, our use of Umwelt is a design heuristic, not an empirical claim about the beetle's experience; we make no claims about the animal's internal or affective states.

8 Conclusion

This paper introduced the Contact Membrane, an interface for real-time listening with a single mealworm beetle, together with Umwelt-centric design, the methodology that produced it. Rather than rendering the insect legible or simulating its point of view, the system stays with a living individual and translates its substrate vibrations into a shared, spatialized soundscape. Across three public performances, the work's central finding emerged from its aesthetics: the soundscape drew audiences in, yet that same beauty led some to doubt whether it masked the insect's voice. Umwelt-centric design functions as a speculative method for surfacing the politics of interspecies listening: who is heard, on whose terms, and at what cost. We offer these findings as a way to keep this politics perceptible.

9 Ethical Standards

While the political implications of translating between Umwelten constitute the conceptual focus of this research, the physical handling of a living organism necessitates a strict, baseline ethical framework. We position care and Haraway's "response-ability" [Haraway 2013] as the procedural philosophy governing the physical interaction itself. Consequently, the biological welfare of the mealworm beetle strictly supersedes any aesthetic or performative goals. To ensure physical well-being and agency, we implemented the following matter-of-fact protocols:

(1) **Voluntary Participation (Assent):** The insect is never physically restrained, chemically sedated, or glued. The system design allows the beetle to express spontaneous behavior. If the insect attempts to leave the membrane repeatedly, this is interpreted as a withdrawal of assent, and the performance is immediately terminated.
(2) **Minimization of Stress (Acoustic & Tactile):** We carefully monitored the environment to avoid distressing stimuli. The "Contact Membrane" uses feedback from the beetle's own movement, ensuring the acoustic and vibrational levels remain within a natural, non-threatening range for the species. The granular synthesis processing is designed to be audible to the human audience without creating excessive feedback loops that would cause tactile stress to the insect on the paper surface.
(3) **Post-Performance Care:** Interaction duration is strictly limited (under 10 minutes) to prevent exhaustion. Following the performance, the beetle is returned to a food-rich, temperature-controlled habitat, ensuring a quality of life consistent with pre-study conditions.

10 Why it matters

More-than-human art often has a quiet assumption: that the right technology finally lets us hear what a subject "really" feels. Contact Membrane refuses that promise. It treats the gap between species not as a problem to engineer away, but as the thing to make audible, staging its own mediation so that listeners hear a curation. For anyone designing with living organisms, Umwelt-centric design offers a usable stance: build the interface that lets the other remain other, and turn the politics of who-gets-heard from a hidden default into the subject of the work.

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